Name _____ Period _____ Date _____

4-2 Lesson Quiz: Graphing Rational Functions

short form

Question numbers match the Savvas Lesson Quiz. Questions 2 and 4 are not on this form.

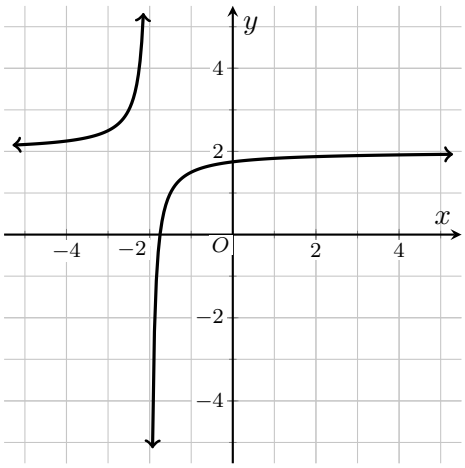
1. Which equation is shown in the graph?

Ⓐ $p(x) = 2 - \dfrac{1}{2x-4}$

Ⓑ $p(x) = 2 + \dfrac{1}{2x+4}$

Ⓒ $p(x) = 2 - \dfrac{1}{2x+4}$

Ⓓ $p(x) = 2 + \dfrac{1}{2x-4}$



The asymptotes I see on the graph: $x =$ _____ $y =$ _____

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3. What are the vertical and horizontal asymptotes of the graph of $f(x) = \dfrac{3-2x}{4-3x}$?

Ⓐ $x = \frac{4}{3}$ and $y = \frac{2}{3}$

Ⓑ $x = \frac{2}{3}$ and $y = \frac{4}{3}$

Ⓒ $x = \frac{3}{4}$ and $y = \frac{2}{3}$

Ⓓ $x = \frac{3}{2}$ and $y = \frac{4}{3}$

Show your work.

Vertical asymptote (where is the denominator 0?):

_____

Horizontal asymptote (compare degrees, then leading coefficients):

_____

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5. The graph of which equation below has a horizontal asymptote at $y = -3$?

Ⓐ $y = \dfrac{x-3}{x+1}$

Ⓑ $y = \dfrac{x}{2-3x}$

Ⓒ $y = \dfrac{1-3x}{2+x}$

Ⓓ $y = \dfrac{1+3x}{2-3x}$

Write the horizontal asymptote of each choice.

A: $y =$ _____ B: $y =$ _____

C: $y =$ _____ D: $y =$ _____

Explain in one sentence how you found them.

_____

Scoring: Question 3 and Question 5 are 10 points each. Question 1 is a bonus of up to 5 points. Total out of 20.

4-2 Lesson Quiz, short form: answer key

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1. C, $p(x) = 2 - \dfrac{1}{2x+4}$**.** The graph has vertical asymptote $x = -2$ and horizontal asymptote $y = 2$, so the denominator is $2x+4$ (rules out A and D). The right branch is *below* $y = 2$: at $x = 0$, C gives $2 - \frac{1}{4} = 1.75$ and B gives 2.25. Savvas quiz item 1, DOK 1, Realize skill F20. Taught by Example 4 (asymptotes, then a test point). **Bonus 5:** 2 for both asymptotes written correctly, 3 for C.

3. A, $x = \frac{4}{3}$ **and** $y = \frac{2}{3}$**.** $4 - 3x = 0$ gives $x = \frac{4}{3}$ (the numerator is $3 - 2 \cdot \frac{4}{3} = \frac{1}{3} \neq 0$). Same degree, so $y = \frac{-2}{-3} = \frac{2}{3}$. Savvas quiz item 3, DOK 1, Realize skill F76. Taught by Example 4. **10 points:** 6 for A, 2 for the vertical-asymptote work, 2 for the horizontal-asymptote work. Distractor B swaps the two; C solves $4x - 3 = 0$; D uses the numerator.

5. C, $y = \dfrac{1-3x}{2+x}$**.** Every choice has equal degrees, so the asymptote is the ratio of leading coefficients: A $\frac{1}{1} = 1$; B $\frac{1}{-3} = -\frac{1}{3}$; C $\frac{-3}{1} = -3$; D $\frac{3}{-3} = -1$. Savvas quiz item 5, DOK 2, Realize skill F76. Taught by Example 3 (Case 3). **10 points:** 6 for C, 3 for the four asymptotes (any three correct), 1 for the sentence naming degree and leading coefficients.

Items left off this form (available for a retake or Form B): Savvas item 2, vertical asymptotes of $y = \frac{(x+1)^2}{x^2-3x-4}$ (answer D, $x = 4$ only: $x^2 - 3x - 4 = (x-4)(x+1)$ and $x+1$ is also a factor of the numerator, so $x = -1$ is not an asymptote; DOK 2, F76; taught by Examples 2 and 6, both cut); Savvas item 4, water added to 10 liters of a 12% salt mixture, $f(x) = \frac{1.2}{x+10}$, liters of water for a 1% mixture (answer 110 liters: $\frac{1.2}{x+10} = 0.01$ gives $x + 10 = 120$; DOK 2, F74; taught by Example 5, cut; the packet's Practice block is its parallel). The TE prints item 4's function as a "percent" but it gives a fraction (0.01 for 1%).

IXL replacement: SmartScore 70 or above on 7JJ and 79Q replaces this quiz with 20/20.